Sentiment Analysis of Tweets using the Sentiment140 Dataset

Background

With the rapid growth of social media, platforms like Twitter have become a vital source of public opinion on products, events, and topics. Analysing this data can provide businesses, researchers, and policymakers with valuable insights into public sentiment and emerging trends.

The **Sentiment140 dataset**, containing 1.6 million labelled tweets, offers an opportunity to build and evaluate models that can automatically classify the sentiment of tweets as **positive**, **neutral**, or **negative**.

Objective

To develop a **machine learning-based sentiment analysis model** capable of classifying tweets into one of three categories — **positive (4)**, **neutral (2)**, or **negative (0)** — based on their textual content.

Goals

1. **Data Preprocessing:**
   Clean and preprocess raw tweet text by removing noise such as URLs, mentions, hashtags, and special characters.
2. **Exploratory Data Analysis (EDA):**
   Analyse sentiment distribution, frequently used words, and linguistic patterns in positive and negative tweets.
3. **Feature Engineering:**
   Convert tweets into numerical features using techniques like **TF-IDF**, **CountVectorizer**, or **word embeddings**.
4. **Model Development:**
   Build and evaluate machine learning models (e.g., Logistic Regression, Naive Bayes, Random Forest) to classify sentiments.
   - Optional: Compare with a deep learning model (e.g., **LSTM** or **BERT**) for advanced performance.
5. **Model Evaluation:**
   Assess model performance using metrics such as **accuracy**, **precision**, **recall**, **F1-score**, and **confusion matrix**.
6. **Visualization & Insights:**
   - Visualize sentiment trends and common keywords.
   - Provide insights on how sentiment varies across topics or keywords.
7. **(Optional Enhancement)**
   Deploy a **Streamlit web app** where users can input a tweet and view the predicted sentiment in real time.

Expected Outcome

A trained sentiment analysis model capable of accurately classifying unseen tweets into **positive**, **neutral**, or **negative** categories.
The project will demonstrate the ability to:

- Process and clean unstructured text data,
- Build and evaluate supervised learning models, and
- Communicate findings through meaningful visualizations or an interactive application.

Real-World Relevance

- **Businesses:** Understand customer feedback and brand perception.
- **Governments:** Monitor public opinion on policies.
- **Media:** Track sentiment trends on current events.
- **Researchers:** Study linguistic patterns and emotional expression online.

PROJECT SUBMISSION LINK:

https://docs.google.com/spreadsheets/d/1kAeb1h4e-xqS0jR4teNzRJBvOAQa3fVUZ8WmzX5cFy4/edit?usp=sharing